Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Ustaad उस्ताद

The senior technician in every repair worker's pocket

Team name: SomilBuilders
Team leader name: Kaustubh Upadhyay

Problem Statement: Future of Work & Enterprise Productivity. India's AC repair technicians learn on the job and get stuck alone at the customer's door. Ustaad gives them a senior's help by voice, in their language, safely.

SOMILBUILDERS

Kaustubh Upadhyay
Voice & agent

Samyak Jain
Knowledge & commerce

Brief about the idea

Talk through a repair. Get the next safe step.

Raju fixes ACs in Lucknow. When a unit blinks E6 and he is stuck, he calls a busy senior or searches a backlog of WhatsApp voice notes.

Ustaad is a voice assistant on his phone. He asks in Hindi. Ustaad answers out loud, one step at a time, using fixes his group already shared—and stops him before anything dangerous.

Hindi | **Telugu** | **Tamil** | **Bengali** | **English**

Voice in > backed by real fixes > safety check > next step



A real answer: the fix Ramesh shared in May 2025

Opportunities

Millions more ACs, not enough trained hands

66-345 million room ACs in India, 2023 to 2037

CEEW 2025

5× more skilled AC technicians needed

0.37 M to 2 M · CEEW 2025

4 in 10 technicians have no formal training

CEEW 2023 · n = 369

How different is it from any of the other existing ideas?

Today's option	Voice	Hands-free	Own fixes	Safety	Parts
Call a senior	Yes	-	Yes	sometimes	-
YouTube videos	sometimes	-	-	-	-
Generic AI chat	Yes	-	-	-	-
Ustaad	**Yes**	**Yes**	**Yes**	**Yes, in code**	**Yes**

How will it be able to solve the problem?

It puts a senior's knowledge on the worker's phone when he is stuck: spoken step by step, and backed by fixes his group recorded.

USP of the proposed solution

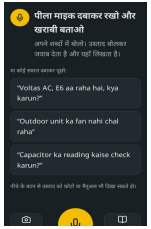
Group voice notes become credited repair knowledge. Code checks safety. Ustaad checks stock and nearby quotes.

List of features offered by the solution

Six things Ustaad does in one repair

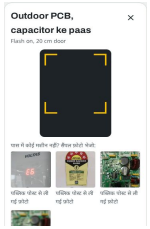
1 Talk, don't type

Speak in five languages. Hear the answer and read captions.



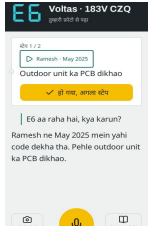
2 Show, don't describe

Send the nameplate, error code or board when Ustaad asks.



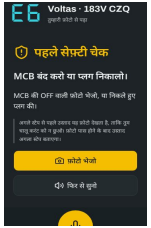
3 Answers you can trust

See who shared the fix and when, or the manual page.



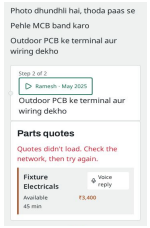
4 Safety first

Show power is off before electrical work. Gas work stops.



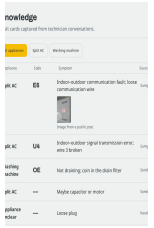
5 Gets the part

Check stock, compare shop quotes and approve.



6 Learns from the group

Turn group voice notes into credited repair cards.

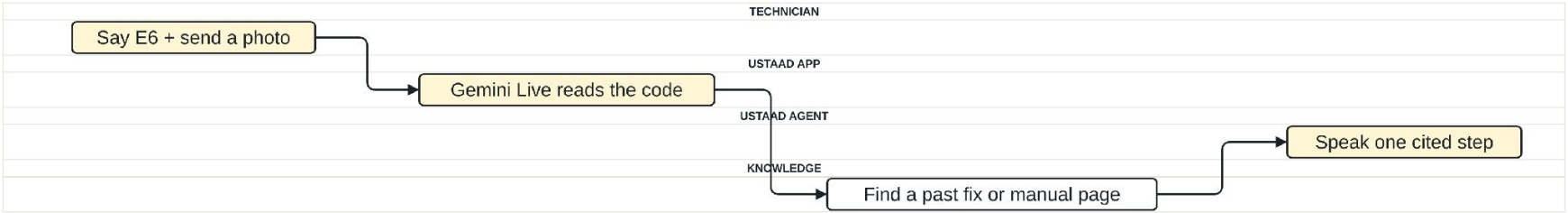


Also remembers each technician: language, brands and what they want to learn (Memory Bank).

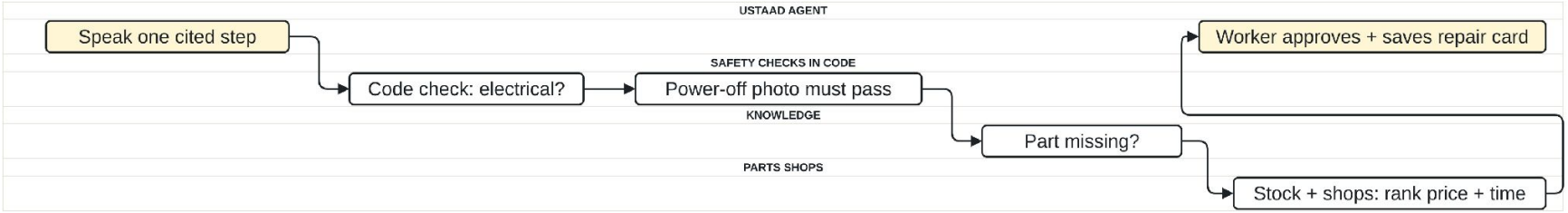
Process flow diagram or Use-case diagram

One repair, from "E6" to fixed

1 ASK > UNDERSTAND > ANSWER



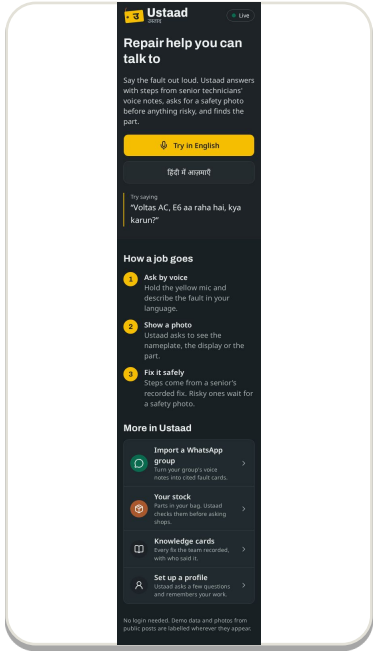
2 CHECK SAFETY > FIND THE PART > FINISH



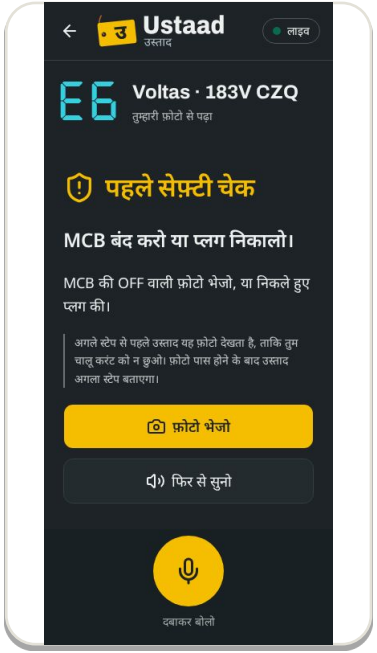
If any check fails, Ustaad asks again. It never skips ahead.

Wireframes/Mock diagrams of the proposed solution (optional)

Designed for one thumb, dirty hands and bright sun



Big yellow mic. Try in English or Hindi. No login.



Safety check: switch off the MCB and send a photo.



Profile and controls in Telugu.

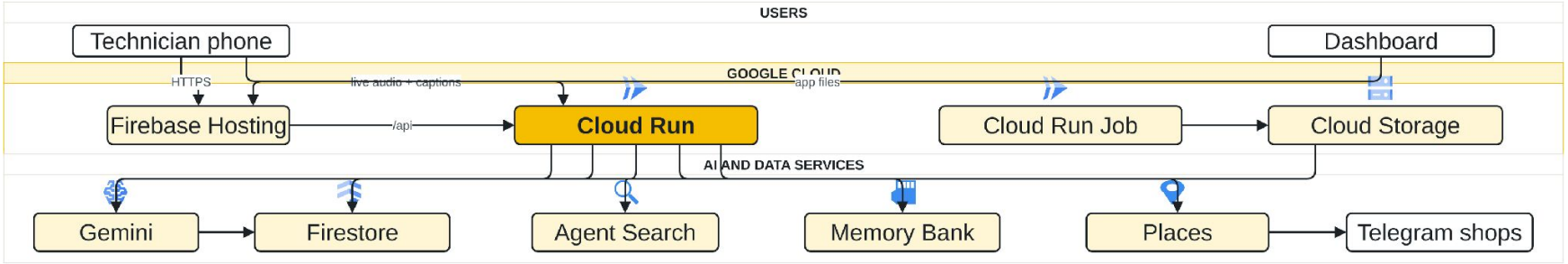
MADE FOR THE JOB SITE

- one step at a time
- who shared the fix, and when
- the error code as the AC displays it
- a clear Stop button

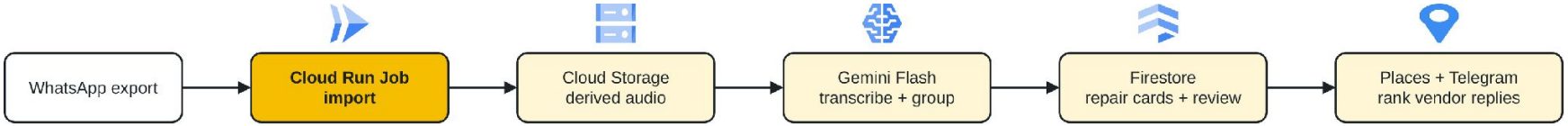
Architecture diagram of the proposed solution

How it's built on Google Cloud

LIVE REPAIR



KNOWLEDGE IMPORT + DASHBOARD



Firebase · Cloud Run · Firestore · Gemini | Voice uses a direct WebSocket to Cloud Run because Hosting rewrites do not carry WebSockets.

Technologies to be used in the solution

Gemini does the thinking. Code does the checking.

Built on Google Cloud

Gemini Live Voice conversation in Hindi, Telugu, Tamil, Bengali and English	**Gemini Flash** Reads photos and turns voice notes into repair cards	**Gemini Embeddings** Finds the right past fix	**Cloud Run + Jobs** API, live voice bridge and WhatsApp import	**Firestore** Repair cards, workers, jobs, stock and quotes
Firebase Hosting Installable phone app at ustaad101.web.app	**Agent Search** Brand manuals cited to the page	**Memory Bank** Remembers the worker across visits	**Cloud Storage** Photos and voice notes	**ADK** The agent and its tools

React + TypeScript PWA · FastAPI · Places API for nearby shops · Telegram for demo vendor messages · WhatsApp Business is the production path after Meta verification

Estimated implementation cost (optional)

Prototype runs on approved Google Cloud credits

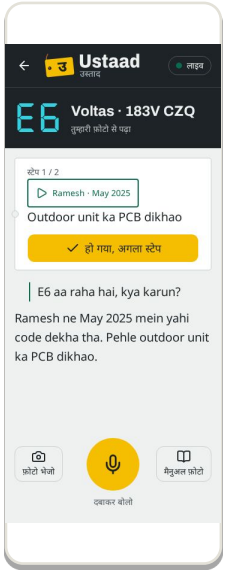
Cost figures are omitted

Not estimated for the prototype. Running on Google Cloud credits.

A production estimate will use measured Cloud Billing reports and real repair-session volume—not a guessed cost per conversation.

Snapshots of the prototype

The working prototype



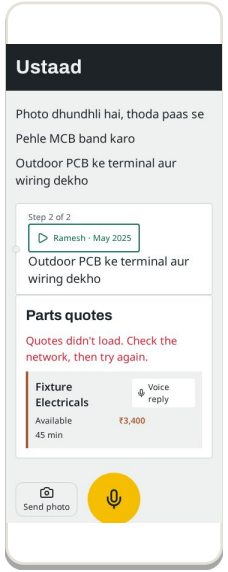
Cited repair step



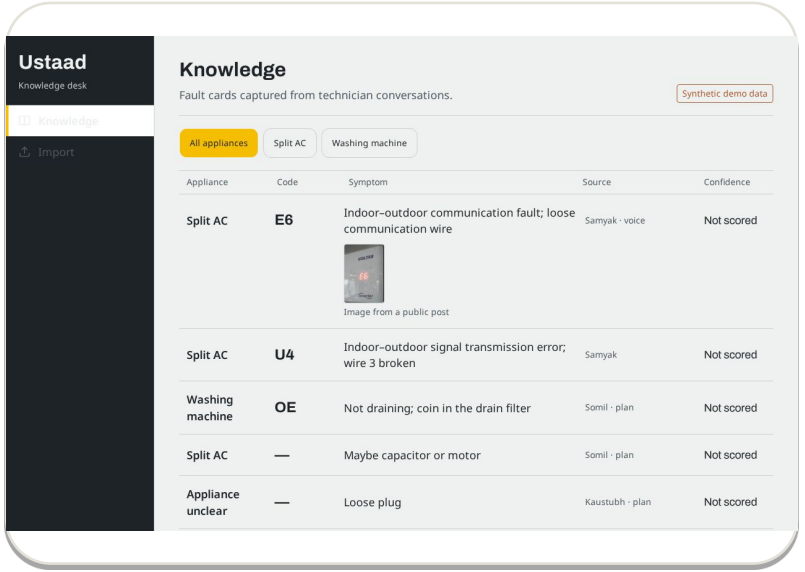
Outdoor PCB photo



Power-off safety gate



Parts quotes



WhatsApp knowledge cards with speaker, date and image provenance

Demo voice notes are team recordings. Demo photos are AI-generated or from public posts, and labelled in the app.

Prototype Performance report/Benchmarking

Tested on real voices, manuals and photos

0 of 10 PASS
ways to skip a safety check worked
Automated code checks

10 of 11
repair cards correct from 20 voice notes
11 of 11 credited · 3 min 20 s

~1 s
until Ustaad starts speaking
0.9–1.1 s · Wi-Fi laptop test

2.4–3.2 s
to find the right manual page
Voltas · Daikin · LG

9 of 9 PASS
voice onboarding checks in Hindi and Telugu
Real Gemini Live model

6 of 6 PASS
test photos identified correctly
4 of 5 own photos · no guess on blur

Not yet measured: reply time on 4G and the full parts-quote round trip. Details: docs/STATUS.md.

Additional Details/Future Development (if any)

From one WhatsApp group to technicians across India

1 Now

AC repair with Voltas, Daikin and LG manuals; Hindi, Telugu, Tamil, Bengali and English; one technician group.

2 Near term

Pilot with technician groups and shops; WhatsApp Business for vendor messages; more manuals; make spoken guidance wait for every safety check.

3 Later

Fridges, washing machines and more trades; fault patterns across nearby pincodes; support for technician certification.

CEEW estimates India needs 5× more skilled AC technicians by 2037. Ustaad helps technicians learn on every job.

Provide links to your:

Try Ustaad on your phone

GitHub Public Repository https://github.com/KaustubhUp025/ustaad

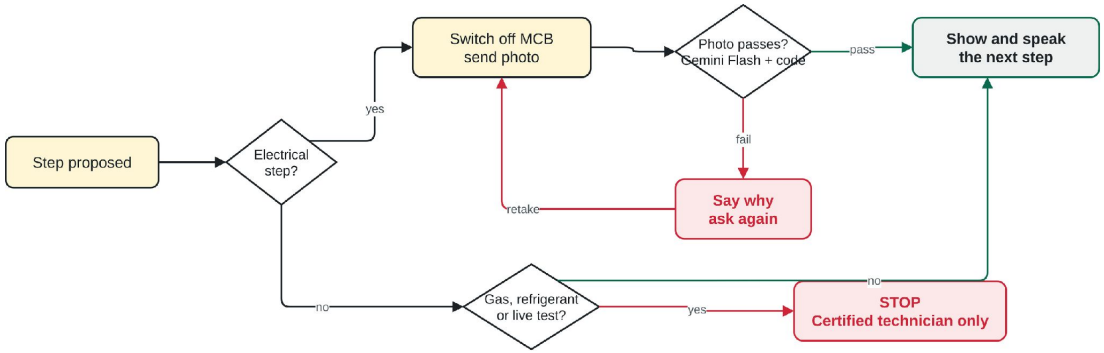
Demo Video Link (3–4 minutes)
https://youtu.be/UyKmPl6wMFU

Final Product Link https://ustaad101.web.app

Open the product link > Try the demo > hold the mic and say: E6 aa raha hai, kya karun?

Safety

Safety is checked in code, not left to the AI



FOUR RULES

- No electrical step until a photo shows power is off
- Gas or refrigerant work stops for a certified technician
- A wrong photo fails, and Ustaad says why
- Stop cancels the repair plan

0 of 10 unsafe-bypass tests succeeded

Sources: CEEW 2023 technician survey · CEEW 2025 workforce-demand study

ceew.in/publications/india-cooling-action-plan-and-air-conditioning-service-sector-perceptions

ceew.in/publications/air-conditioner-technician-demand-and-Sustainability

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

